JNTUH USED 22-11-17PM Set 2

Code No: 117CF **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, November/December - 2017
DESIGN PATTERNS
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART- A

(25 Marks)

1.a) What is design pattern? [2]
b) Explain design patterns in Smalltalk MVC. [3]
c) Draw the structure Factory Method. [2]
d) Explain about recursive composition. [3]
e) Write intent of Proxy pattern. [2]
f) Write motivation of Wrapper pattern. [3]
g) What are known uses of Iterator pattern? [2]
h) Write benefits and drawbacks of Mediator pattern. [3]
i) What is the intent of Strategy pattern? [2]
j) Describe patterns in software. [3]

PART-B

(50 Marks)

2.a) What are some common causes of redesign? Explain.
b) How to use a design pattern? Explain in detail. [5+5]

OR

3.a) Discuss about toolkits and frameworks.
b) Draw a diagram to describe design pattern relationships. [5+5]

4.a) Describe formatting in Lexi's design.
b) What are consequences of Builder pattern? Explain Builder pattern implementation. [5+5]

OR

5.a) Explain about encapsulating the analysis in Lexi's design.
b) Write sample code of Prototype pattern. [5+5]

6.a) Explain the sample code of Flyweight pattern.
b) Discuss about implementation issues of Decorator pattern. [5+5]

OR

7.a) Write about motivation and consequences of Façade pattern.
b) Write sample code of Composite pattern. [5+5]

8.a) Explain about Mediator pattern.
b) Explain about motivation of Mediator pattern. [5+5]

OR

9.a) Briefly discuss about Memento pattern.
b) Describe implementation of Command pattern. [5+5]

10.a) Write some of the benefits and liabilities of the Visitor pattern.
b) Discuss about the pattern community. [5+5]

OR

11. Explain the following:
a) Discussion of behavioral patterns
b) Template method pattern. [5+5]

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